

UNIT 7



SCRATCH



SCRATCH

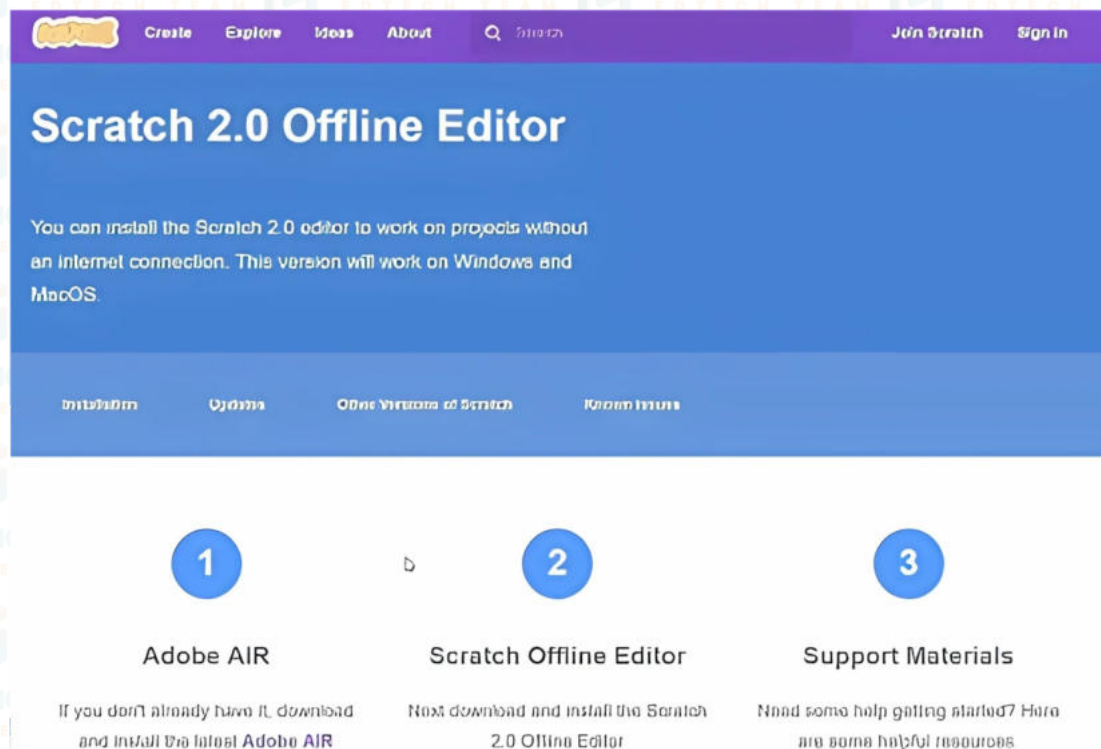
Scratch, a visual programming language developed by the Massachusetts Institute of Technology (MIT) Media Lab. Scratch is designed to be easy to use for students with little to no programming experience. It allows users to create programs by dragging and dropping coloured blocks, rather than writing complex code.

Here are the key points from the text about Scratch:

- Scratch is a free, beginner-friendly programming language.
- Programs in Scratch can include games, animations, interactive stories, music, and art.
- Scratch is a great tool for students to learn about mathematical and computing concepts, creative thinking, logical reasoning, problem-solving, and collaboration skills.
- Scratch can be used online or downloaded for offline use on a computer or Android device.

DOWNLOADING AND INSTALLING SCRATCH OFFLINE

You can download and install Scratch on a computer or an Android device to work offline. The latest version of Scratch can be downloaded for Windows 10 or later versions. However, Scratch 2.0 is a good version that works on most of the computers. Scratch 2.0 can be downloaded from this website: <https://scratch.mit.edu/download/scratch2>

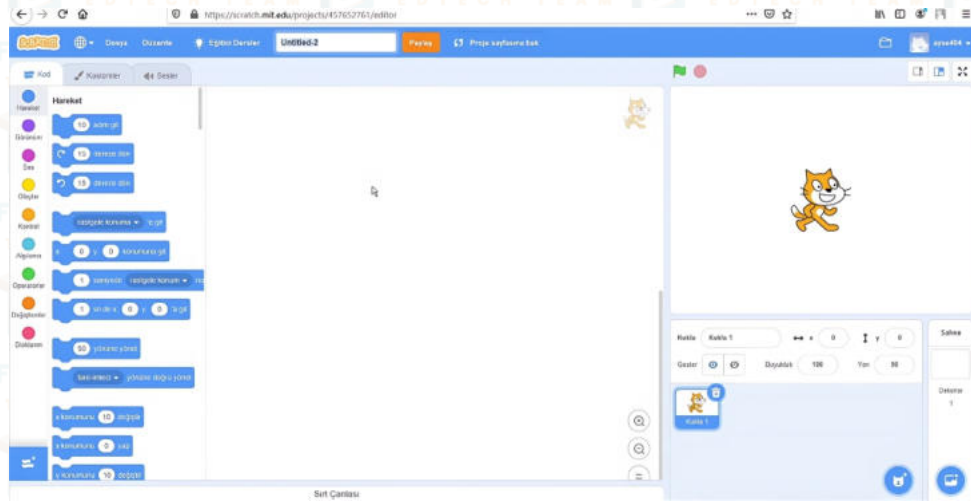


To begin, **Scratch Offline Editor** for Windows needs to be installed on to your computer. Installation of Scratch is very easy.

1. After downloading, we need to just run the executable (.exe) file.
2. The first screen will appear which will ask the location where we want to install the Scratch.
3. It will also ask about the shortcuts to create. You need to just continue with default options.
4. This will initiate the installation process and after installation, Scratch will be ready to use.

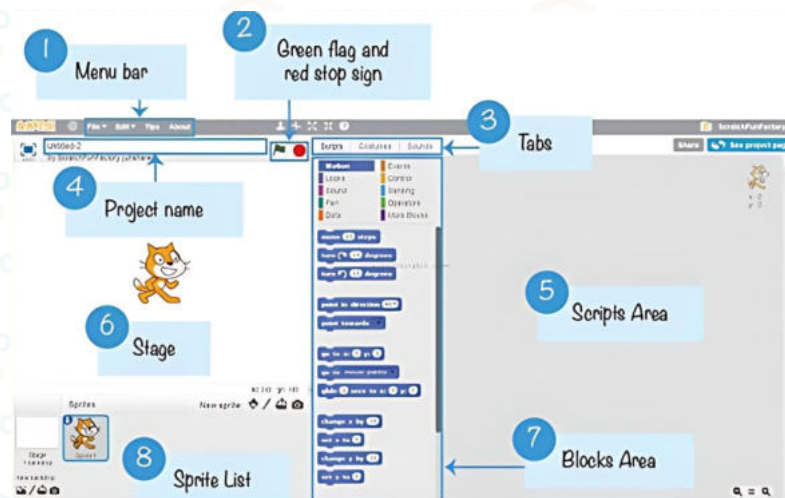
SCRATCH ONLINE

Scratch can also be used online by simply loading its editor in our web browser which is available at: <https://scratch.mit.edu/projects/editor>



(BASICS OF SCRATCH EDITOR)

Scratch Editor:



Stage or Stage Preview Window:

This is like a playground where your characters (sprites) come to life. You can see what your program is doing here, as the sprites move and act according to your code. The stage uses X and Y coordinates to pinpoint where things are

Script Area: This is where you write the instructions (code) that tell your sprites what to do on the stage. Imagine it as giving orders to your characters like "move right 10 steps" or "say hello."

Sprite List: This shows all the characters (sprites) you can use in your project. You can click on them to change their names or how they behave.

Backdrop: This is the background image for your project, like the setting for your characters.

Script Tab (Code Tab): This is where you write the instructions (code) that tell your sprites what to do. You drag and drop coloured blocks together to build your code.

The following table describes the purpose of each block.

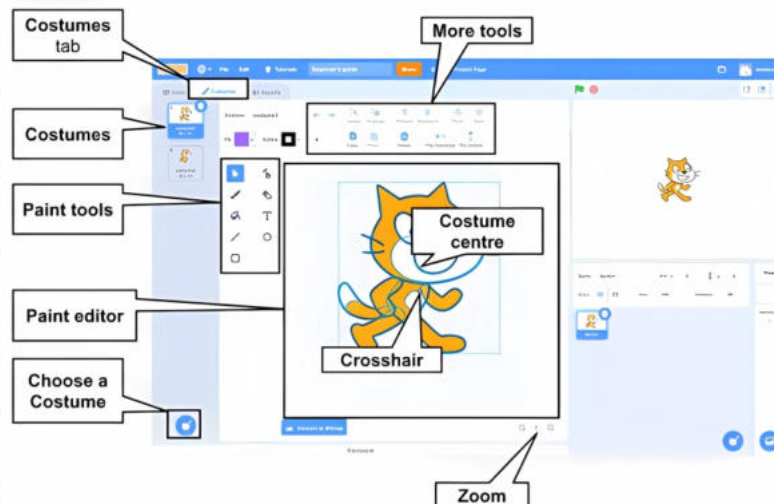
Feature	Description
Blocks	Scratch uses colourful blocks to represent different coding instructions. Snapping these blocks together lets you create programs.
Script Area	This is the designated workspace where you drag and drop blocks to build your program.
Categories	Blocks are grouped into categories based on their function. Examples are Motion, Looks, and Sound.

Palette	The palette located on the left side of the Scratch interface contains all the available block categories.
Search Bar	You can use the search bar to find specific blocks by name.

COSTUMES TAB

In simpler terms, a sprite is like an actor in your Scratch project. You can change how the sprite looks by choosing from different costumes, or you can even design your own costumes using the built-in editor.

1. you can change the appearance of a sprite by clicking on the "Costumes" tab.
2. This tab will show you all the available costumes for the selected sprite.
3. You can also create new costumes for your sprites using the Scratch paint editor.



SOUND TAB

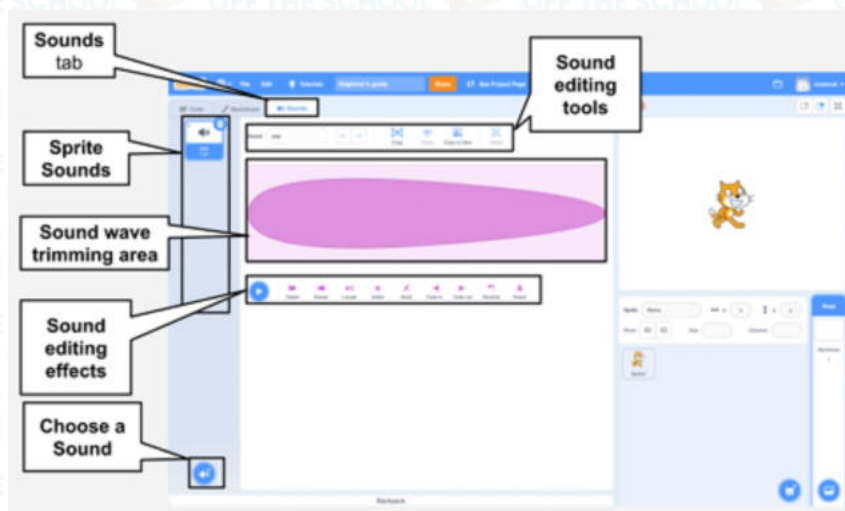
In simpler terms, not all sprites (characters) in Scratch need sounds. The Sound Tab lets you add sounds to your sprites, edit them, or remove them altogether. You can also control how these sounds are played in your project.

Some sprites additionally have at least one sound.

Unlike costumes, sounds are not required for a sprite.

The sounds tab allows you to add, delete, and edit sounds.

You can use the sound editor or blocks to play these sounds in your Scratch project.

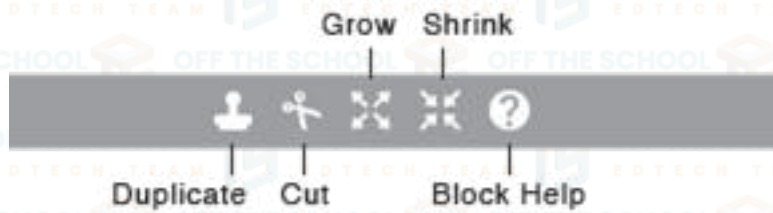


CURSOR TOOL

This toolbar has five buttons that let you manipulate sprites (characters) in your project:

- **Duplicate:** Creates a copy of the selected sprite.
- **Delete:** Removes the selected sprite.

- **Grow:** Makes the selected sprite larger.
- **Shrink:** Makes the selected sprite smaller.
- **Help:** Provides information about using Scratch (not shown in the image).



BASICS ABOUT CREATING PROGRAMS

Few points should be kept in mind before developing a project.

Most of the Scratch programs contain Sprites, Backdrops and Code Blocks.

Every Sprite in a program has a separate Code Block which controls its actions.

You can develop programs for different logics like storytelling, sprite animation, simple game and others.

Open Scratch Editor:

From Events Option in Script, drag and drop on script area.

From Control Option, drag forever and drop on script area. Inside the forever block, drag and drop the illustrated commands and change values accordingly. Colors will help you to find the option after completing the codes blocks, play (run) and stop the program. After completing the codes blocks, Play (Run) and Stop the program.

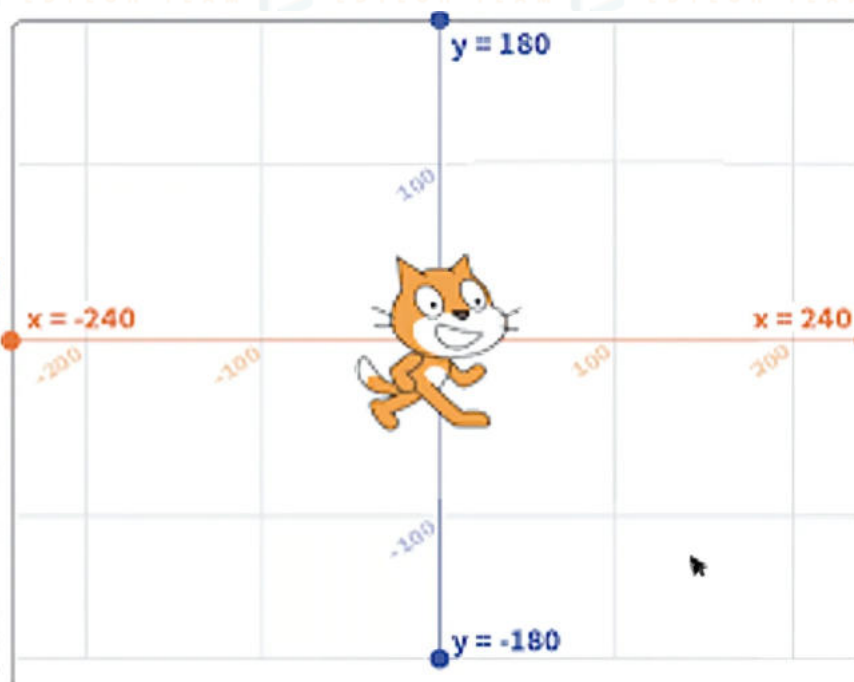
Playing and Stopping the Animation/Program:

The Green Flag icon starts your program, letting you see your animation or code in action.

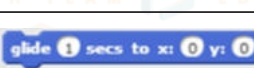
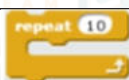
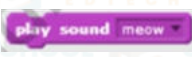
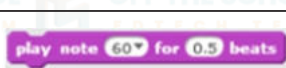
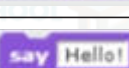
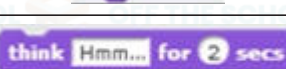
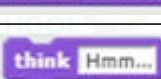
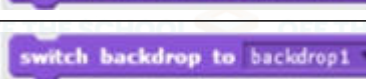
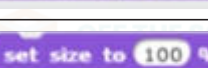
The Red Stop icon stops your program if it's running.

Coordinates on Scratch Stage:

Scratch uses a 2D coordinate system to determine the location of sprites on the stage. The X coordinate represents the horizontal position, while the Y coordinate represents the vertical position. The center of the stage is (0, 0), with positive X values moving to the right and positive Y values moving up. The X coordinate can range from -240 to 240, and the Y coordinate can range from -180 to 180.



SOME IMPORTANT COMMANDS IN SCRATCH

Feature	Group	Description
	Event	It triggers the following code blocks when the flag is clicked.
		It triggers the following code blocks when a specific key is pressed.
	Motion	Moves the sprite in the current direction for specified steps.
		Turns the sprite to the specified degree.
		Sends sprite to specified x, y coordinates (position on stage).
		Sprite glides to specified x, y coordinates (position on stage).
	Control	Waits for specified seconds.
		Repeats the following code blocks for the specified number of times.
		Keeps repeating the following code blocks for infinite times.
		Executes the block when the given condition is true.
	Sound	Plays the specified sound.
		Plays the specified drum sound for specified beats.
		Plays the specified note for specified beats.
		Changes the musical instrument.
	Looks	Says the given words for specified seconds.
		Says the given phrase.
		Pretends to think by showing given phrase for specified seconds.
		Pretends to think by showing, given phrase.
		Changes the costume to the selected value.
		Changes the background to the selected value.
		Changes the size of Sprite.
	Sound	Pen is down. Now a line is drawn as Sprite moves.
		Pen is up.
		Sets the color of the pen to draw line.

After learning some frequently used commands, let's develop few programs. Try developing the illustrated program.

1. CHANGING THE COSTUME

With this small set of code blocks the costume of the spirit changes and it looks as if the spirit is moving. Can you explain this program.

2. ADDING AND MOVING SPRITE

For the next program, you should right-click on the Sprite (Cat) and delete it. Now you don't have any Sprite in your project.

To add a Sprite, click on "Choose sprite from library". This will open a dialogue box showing all available Sprites. Select Beetle from the list.

On the Stage, Move the Beetle to the Upper-left corner.

Now add the following codes.

3. PLAYING SOUND IN SCRATCH

We can add sounds in our projects. It is very simple and easy. We can play different sounds like beat the drum or play different notes. You can use these sounds to make any animation, game or even just creating music. You can find complete list of notes and beats on following link: <https://en.scratch-wiki.info/>

- In this sample program, notes are used to create the music of a song.
- Try to develop this program and write the output of this program.

4. CHANGING THE BACKDROPS

By default, the Background of stage is white/ blank. It can be changed. Place the cursor on "Choose a Backdrop" button and click. It consists of four different options:

• Choose a backdrop from library:

You can choose a backdrop from available library of backdrops.

• Paint New backdrop:

You can also paint a new backdrop

• Upload backdrop from file:

An image on your computer can also be used as backdrop.

5. NEW BACKDROP FROM CAMERA

To capture a backdrop in Scratch and then use it in your project. It also provides a step-by-step guide to creating a program that changes the backdrop based on a sprite's movement. You'll add two backdrops (Blue Sky and Desert) and a dinosaur sprite, then code the dinosaur to switch the backdrop to Desert when it reaches a certain point on the stage.

6. USING MULTIPLE SPRITES

There are four ways to add sprites:

- Choose a sprite from the library
- Paint a new sprite
- Upload a sprite from a file
- Create a new sprite from your webcam

7. TAKING INPUT

Scratch also allows developers to take input from the user. This illustration shows a simple program to take the input. In this program, you can use an ask command in which you may write your message. Like in this example, the name of the user is asked. The third line

is important and you need to add a say command and join the answer command with this. The input that you take in ask is stored in answer and it is displayed using say. By understanding the logic of this program, now you can calculate many other things. Remember, developing programs is great activity and coding is going to be an important skill in the future. Through some basic skills that you learned in this unit, you can develop many complex projects by using your creativity and imagination.